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Research papers

RUL-Mamba: Mamba-based remaining useful life prediction for lithium-ion batteries

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A R T I C L E I N F O

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A B S T R A C T

Lithium-ion batteries play a crucial role in the fields of renewable energy and electric vehicles. Accurately predicting their Remaining Useful Life (RUL) is essential for ensuring safe and reliable operation. However, achieving precise RUL predictions poses significant challenges due to the complexities of degradation mechanisms and the impact of operational noise, particularly the capacity regeneration phenomenon. To address these issues, we propose a lithium-ion battery RUL prediction model named RUL-Mamba, which is based on the Mamba-Feature Attention Network (FAN)-Gated Residual Network (GRN). This model employs an encoder-decoder architecture that effectively integrates the Mamba module, FAN network, and GRN network. Mamba demonstrates superior temporal representation capabilities alongside efficient inference properties. The constructed FAN network leverages a feature attention mechanism to efficiently extract key features at each time step, enabling the Mamba block in the encoder to effectively capture information related to capacity regeneration from historical capacity sequences. The designed GRN network adaptively processes the decoded features output by the Mamba blocks in the decoder through a gating mechanism, accurately modeling the nonlinear mapping relationship between the decoded feature vector and the prediction target. Compared to state-of-the-art (SOTA) time series forecasting models on three battery degradation datasets from NASA, Oxford and Tongji University, the proposed model not only achieves SOTA predictive performance across various prediction starting points, with a maximum accuracy improvement of 42.5 % over existing models, but also offers advantages such as efficient training, fast inference and being less influenced by the prediction starting point. The source code and datasets are available at https://github.com/USTC-AI4EEE/RUL-Mamba.

1. Introduction

In the face of the challenges posed by climate change and the energy crisis, the demand for clean and renewable energy continues to grow [1–3]. Lithium-ion batteries, serving as energy storage devices, are widely used in energy storage systems and electric vehicles owing to their remarkable attributes such as high energy density, stable electrochemical properties, long cycle life, and environmental friendliness [4,5]. However, with increased usage, lithium-ion batteries undergo aging, leading to a decline in their health status, which can affect the overall lifespan of the batteries and pose serious safety risks to energy storage systems [6,7]. The remaining useful life (RUL) is a crucial indicator for assessing battery health. Accurately predicting the RUL of lithium-ion batteries can enhance the effective management and planning of energy resources. Meanwhile, it can improve the utilization efficiency of clean energy and avoid the occurrence of serious accidents caused by battery aging [8].

Lithium-ion battery RUL prediction methods can be broadly classified into model-based and data-driven methods, depending on the modeling techniques used [9,10]. Model-based methods include mathematical models, a set of algebraic, empirical equations, and associated parameters, all of which rely on experimentation and large volumes of empirical data. These methods allow for the modeling of battery degradation behavior using either physics-based or regression approaches, enabling accurate extrapolation for predicting battery performance. Model-based methods encompass physics-based models, equivalent circuit models, and empirical models [11]. Physics-based models [12,13] are constructed based on the internal physical and

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electrochemical phenomena of batteries. By utilizing theories such as reaction kinetics and electrode porosity, these models help in understanding the physicochemical processes within batteries and estimating their state. Although physics-based models provide clear mechanisms and strong interpretability, they exhibit high computational complexity, sensitivity to parameters, and poor real-time performance, making them difficult to directly apply for online RUL prediction. Equivalent circuit models [14,15] describe the degradation mechanisms of batteries using circuit analysis techniques, allowing for the simulation of important parameters under various operating conditions. The parameters of these models can be determined through electrochemical impedance spectroscopy (EIS) datasets, enabling RUL predictions. While equivalent circuit models are computationally efficient and easy to integrate with Battery Management Systems (BMS), they face challenges such as low parameter identification accuracy and limited capability to characterize complex aging mechanisms. Empirical models [16–19] predict battery degradation trends using regression models or empirical formulas, typically based on health indicators like voltage, internal resistance, and temperature. Since these models do not consider the internal characteristics of the battery and mainly rely on historical data, they often incorporate filtering techniques to update model parameters. Although empirical models are simple to use and have a short development cycle, they tend to have poor extrapolation capabilities and weak generalization abilities in unknown operating conditions.

In contrast, data-driven methods do not need to consider the complex electrochemical reactions and failure mechanisms within the battery. Instead, they leverage extensive real measurement data to learn the nonlinear mapping relationship between relevant parameters and future battery capacity for RUL prediction [20]. These methods can be further divided into traditional machine learning methods and deep learning methods [21]. Traditional machine learning methods, such as Support Vector Machines (SVM) [22,23], Support Vector Regression (SVR) [24,25], and Gaussian Process Regression (GPR) [26,27], can extract valuable insights from complex high-dimensional time series data, enabling accurate prediction of battery performance. Nevertheless, despite their simplicity and efficiency in modeling, their capacity to learn nonlinear features is circumscribed. This limitation paves the way for deep learning methods to replace them in RUL prediction tasks [28].

In recent years, deep learning methods have garnered significant attention for their exceptional feature extraction and nonlinear mapping capabilities in the time series analysis of battery systems [29–31]. These methods can be roughly classified into those based on Recurrent Neural Networks (RNNs) class and those based on Transformer [32]. RNN-based methods have demonstrated their effectiveness in handling sequential data when predicting RUL. For instance, researchers have proposed using RNN to model the complex nonlinear trends associated with battery degradation in order to effectively understand capacity changes [33–35]. Long Short-Term Memory (LSTM) networks have been employed to model nonlinear capacity curves for assessing the State of Health (SOH) of batteries [36–38]. Additionally, Gated Recurrent Units (GRUs), which are derived from LSTMs, are frequently used for RUL prediction [39–45]. However, RNN-based methods can incur significant time costs during training due to the cyclical modeling of sequences, and they often suffer from performance degradation due to challenges related to long-term dependencies [46,47]. In contrast, Transformer is well-suited for capturing temporal dependencies in long sequence data and has the capabilities of parallel computation and powerful feature extraction [28]. Consequently, many researchers have applied Transformer-based methods to RUL prediction tasks. For example, to achieve superior RUL prediction performance, Chen et al. [48] devised an innovative approach by combining the Denoising Auto-Encoder (DAE) with the Transformer architecture, thereby integrating the processes of denoising and prediction into a unified framework. Cai et al. [49] proposed a hybrid prediction framework that amalgamates Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN), Transformer, and Deep Neural Network (DNN). This elaborate combination is specifically designed to surmount the challenges associated with limited capacity data, enabling highly accurate RUL prediction even when the available data is scarce. Liu et al. [50] proposed a novel patch-based transformer architecture (PatchFormer) for RUL prediction, which uses a Dual-Patch Attention Network (DPAN) to capture global and local correlations in battery capacity degradation series and combines a Feature Attention Network (FAN) to analyze temporal features. However, Transformer-based methods still face inherent limitations, particularly the time-consuming inference caused by the quadratic computational complexity of attention calculations [51]. Table 1 provides a comprehensive summary of various studies on lithium-ion battery RUL prediction, detailing the types of input variables, classifications, and the prediction methods employed.

Table 1
The summary of selected studies on RUL prediction.

Classification	Input variables	Year	Prediction methods
Model-based methods	Capacity data obtained by state-space estimation	2019	Extended Kalman filter (EKF) [16]
	Capacity data obtained by filter algorithm	2019	Unscented Kalman filter (UKF) [17]
	Capacity data obtained by state-space estimation	2020	Unscented particle filter (UPF) [18]
Traditional machine learning methods	Voltage-dependent health features and a thermal-dependent health feature	2021	Gaussian process regression (GPR) [52]
	Historical capacity	2022	Relevance vector machine (RVM) [53]
	Characteristic voltage (CV), cycle times, and capacity	2023	improved PF-RLS algorithm [54]
Deep learning methods	Historical capacity	2018	LSTM-RNN [37]
	Three health indicators	2022	LSTM-based encoder–decoder [55]
	Historical capacity	2022	CNN-ASTLSTM [56]
	Capacity and four health indicators	2022	Attention-based CNN-BiLSTM [57]
	Historical capacity	2023	TCN-LSTM [58]
	Historical capacity	2023	TCN-GRU-DNN [59]
	Historical capacity	2022	Combined DAE and transformer network (DeTransformer) [48]
	Historical capacity	2023	Denoising transformer-based neural network (DTNN) [60]
	Historical capacity	2023	CEEMDAN-transformer-DNN [49]
	Eight health indicators	2024	MambaLithium [61]
	Historical capacity	2025	PatchFormer [50]

To address the computational inefficiency of Transformer on long sequences, Mamba [62] has been introduced as a novel general sequence foundation model backbone capable of performing effective content-based reasoning. Mamba demonstrates superior temporal representation capabilities alongside efficient inference (5× higher throughput than Transformer) properties, making it widely applicable in the field of time series modeling [62]. For instance, the Mamba4Cast model proposed by Bhethanabhotla et al. [63] is based on the Mamba architecture and inspired by Prior-data Fitted Networks (PFNs), achieving near state-of-the-art (SOTA) results in zero-shot time series forecasting (TSF) tasks while being trained solely on synthetic data, all while maintaining scalability and efficient inference. Liu et al. [64] introduced the MambaCPU network, which employs the Mamba structure to model complex dependencies and global associations among various hardware features, further refining the correlations within and between feature groups using intra- and inter-group attention mechanisms, thereby effectively improving the accuracy of CPU performance prediction. Zhao et al. [65] presented the RS-Mamba model, which efficiently completes dense prediction tasks for Very High Resolution (VHR) remote sensing images

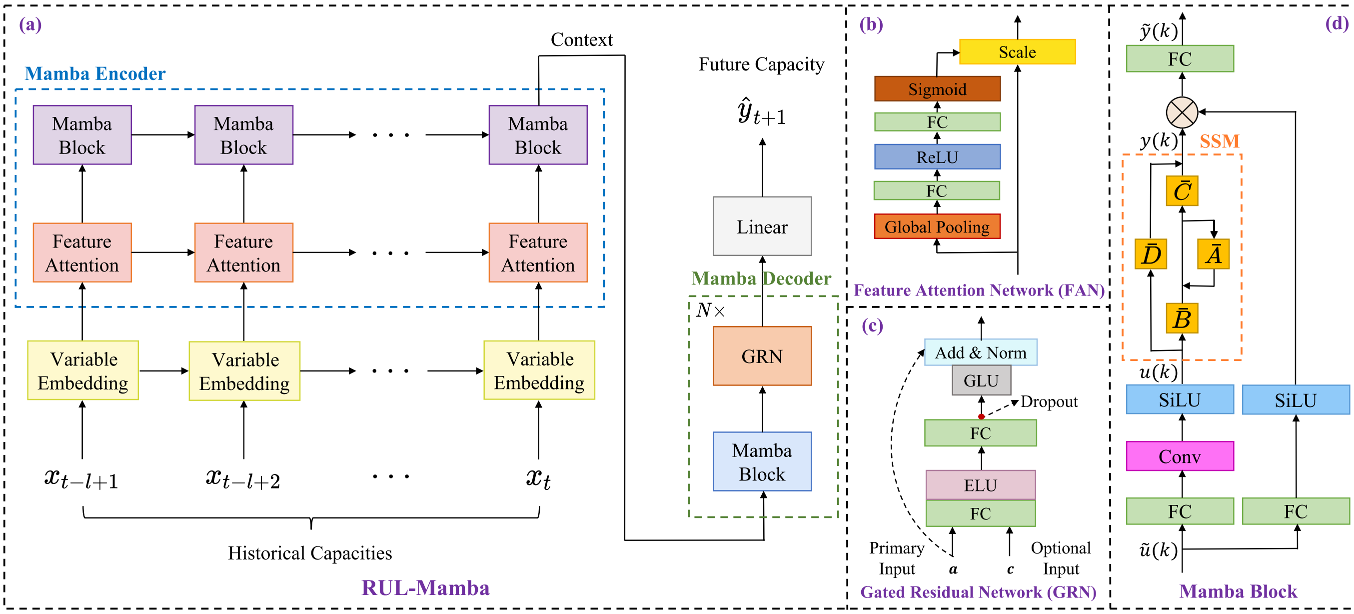


Fig. 1. The network architecture of RUL-Mamba.

by modeling global contextual information and extracting large spatial features in various directions. Cai et al. [66] proposed a new hybrid prediction model to enhance the accuracy of photovoltaic power forecasting, integrating optimized Variational Mode Decomposition (VMD) with Vision Mamba for feature extraction from sky images. The S-Mamba model proposed by Wang et al. [67] replaces the self-attention module in Transformer with bidirectional Mamba blocks to model correlations between variables and employs feedforward networks to capture temporal dependencies, achieving outstanding prediction performance across various TSF tasks in domains such as traffic, weather, and finance, while requiring fewer computational resources.

However, to the best of our knowledge, there has been limited exploration of Mamba in the lithium-ion battery RUL prediction tasks. Recently, Shi [61] proposed the MambaLithium model, which directly applies a pure Mamba network to estimate the RUL of lithium-ion batteries, achieving high estimation accuracy. However, the MambaLithium model estimates the RUL of the battery based on multiple health features in the current state, without fully considering the dynamic changes of the battery's working conditions over time and the possible phenomenon of capacity regeneration. Furthermore, RUL estimation tends to provide relatively accurate information only as the battery approaches the end of its life, lacking the capability for early warning. In contrast, RUL prediction offers stronger early warning capabilities, which is crucial for some critical application scenarios such as aerospace and electric vehicle power systems [58].

In this paper, we propose a lithium-ion battery RUL prediction model (RUL-Mamba) based on Mamba - Feature Attention Network (FAN) - Gated Residual Network (GRN). This model employs an encoder-decoder architecture that effectively integrates the Mamba module, FAN network, and GRN network. The Mamba blocks used in both the encoder and decoder can generate personalized outputs based on specific model inputs through a time-varying selection mechanism. The constructed Mamba encoder, combined with the FAN, utilizes a feature attention mechanism to effectively extract key features at each time step. The designed Mamba decoder integrates the GRN, which selects the appropriate degree of nonlinear processing for feature selection through a gating mechanism. RUL-Mamba not only excels in precisely characterizing the intricate degradation trends within the battery capacity sequence but also showcases an impressive aptitude for effectively capturing local capacity regeneration phenomena in lithium-ion batteries. It achieves a higher level of accuracy in the RUL prediction, providing more reliable forecasts that are of great practical value for battery management and various related applications. The main contributions of this paper are as follows:

(1) This paper proposes a novel hybrid model named RUL-Mamba, which effectively combines the designed FAN network and GRN network with the Mamba module to achieve efficient and accurate battery RUL prediction.
(2) The constructed FAN network extracts relevant input features at each time step through the feature attention mechanism and assigns corresponding weights to enhance the features with important information in each time step, facilitating the Mamba block in the encoder to effectively capture the information related to capacity regeneration in the historical capacity sequence.
(3) The designed GRN network adaptively processes the decoded features output by the Mamba blocks in the decoder through the gating mechanism and accurately models the nonlinear mapping relationship between the decoded feature vectors and the prediction target.
(4) We explore the potential of Mamba in the field of lithium-ion battery RUL prediction with univariate and multivariate inputs. Compared with various SOTA TSF methods, for RUL prediction on three public datasets from NASA, Oxford and Tongji University, RUL-Mamba consistently demonstrates superior performance. Moreover, RUL-Mamba possesses advantages such as efficient training, fast inference and being less influenced by the prediction starting point.

The following sections of this study are organized as follows: Section 2 provides an overview of the details and experimental procedures of the proposed method. Section 3 discusses the datasets used, dataset division, hyperparameter settings, and evaluation metrics. Section 4 presents a comparison of the proposed model with various SOTA methods across three public datasets. Finally, Section 5 summarizes the main conclusions of this study.

2. Method

The overall architecture of the proposed RUL-Mamba model is illustrated in Fig. 1. The RUL-Mamba model adopts an encoder-decoder architecture. The encoder is responsible for processing the historical capacity sequence information, while the decoder processes the intermediate representation generated by the encoder and outputs the prediction result. The encoder is composed of a Feature Attention Network (FAN) and a Mamba block. The decoder is composed of Mamba blocks and Gated Residual Networks (GRNs). The embedding layer transforms the input variable into a high-dimensional space to enable subsequent effective feature extraction. The encoder integrates FAN at the input end of the Mamba block to enhance the model's effective selection of features. The encoder passes the last output of the Mamba block as context to the decoder. The decoder is stacked by N layers composed of Mamba block and GRN, wherein GRN can endow the decoder with flexible nonlinear mapping ability. Finally, the linear layer maps the output of the decoder to the target variable.

2.1. Variable embedding layer

In lithium-ion battery RUL prediction, capacity is often used as the input variable for the prediction model. This paper employs a linear mapping layer to convert a single variable into a D-dimensional embedding vector, which facilitates the subsequent modeling of both time and feature dimensions:

$$X_{emb} = Embedding(X_{in}) \tag{1}$$

where $X_{in} \in \mathbb{R}^{L\times M}$ represents the input of the model, and $X_{emb} \in \mathbb{R}^{L\times D}$ represents the corresponding embedding vector.

2.2. Feature attention network (FAN)

To effectively extract input features, we design the Feature Attention Network (FAN) based on the Squeeze-and-Excitation (SE) [68] mechanism. FAN applies adaptive weighting to the input features, emphasizing the important ones while suppressing less significant features, thereby enhancing the model's performance. The structure of FAN is illustrated in Fig. 1(b) and primarily consists of three operations: Squeeze, Excitation, and Scale.

The squeeze operation uses global average pooling of features to directly compress the $L \times D$ feature map containing global information into a $1 \times D$ feature vector. In essence, this transformation takes each individual one-dimensional feature and condenses it into a single value, endowing it with a global receptive field, so that the correlation between features can be better utilized. The Squeeze operation is defined as follows:

$$Z = F_{sq}(U) = GlobalPooling(U) \tag{2}$$

where the input to FAN is denoted as $U \in \mathbb{R}^{D\times L}$, and $Z \in \mathbb{R}^{D}$ represents the output after the Squeeze operation applied to $U \in \mathbb{R}^{D\times L}$.

The Excitation operation involves passing the feature-wise statistics through a non-linear bottleneck, typically implemented as two fully connected layers with a non-linearity (such as *ReLU*) between them. This allows the model to learn the relative importance of each feature. The Excitation operation is defined as follows:

$$S = F_{ex}(Z, W) = Sigmoid(W_2 ReLU(W_1 Z)) \tag{3}$$

where $S \in \mathbb{R}^{D}$ stands as the output obtained after applying the Excitation operation to $Z \in \mathbb{R}^{D}$, and W_1 and W_2 denote the weight matrices of the respective fully connected layers. Here, $Sigmoid(\cdot)$ and $ReLU(\cdot)$ are both commonly used activation functions.

Finally, the Scale operation implements a recalibration strategy that fine-tunes the original feature map. It accomplishes this by applying the learned scale factors through element-wise multiplication. The Scale operation is defined as follows:

$$\widetilde{U} = F_{scale}(U, S) = U \odot Expand(S, size(U)) \tag{4}$$

where the output of the FAN is denoted as $\widetilde{U} \in \mathbb{R}^{D\times L}$, and the symbol $\odot$ signifies element-wise matrix multiplication. Additionally, the $Expand(\cdot)$ operation refers to the process of expanding the normalized attention weight S to match the dimensions of U.

2.3. Gated residual network (GRN)

Owing to the inability to foresee the precise mapping relationship between the feature vector representation of the historical capacity sequence of lithium-ion batteries and the future capacity, it becomes extremely challenging to ascertain the requisite degree of nonlinear processing for each feature vector. To provide the network with a flexible nonlinear mapping ability, this study constructs a Gated Residual Network (GRN) with a residual structure and introduces a gating mechanism for information selection. The structure of the GRN network is depicted in Fig. 1(c). GRN accepts primary input a and optional input c, the calculation formulas are shown as follows:

$$GRN(a, c) = LayerNorm(a + GLU(\eta_1)) \tag{5}$$

$$\eta_1 = W_1 \eta_2 + b_1 \tag{6}$$

$$\eta_2 = ELU(W_2 a + W_3 c + b_2) \tag{7}$$

where ELU refers to the exponential linear activation unit [69], η_1, η_2 are intermediate layers' results, W_i and b_i represents the weights and biases of linear layers, and LayerNorm is a kind of layer normalization method [70]. Gated Linear Unit (GLU) utilizes the sigmoid function to implement the gate mechanism for information selection. When the information is highly responsive, GLU allows all the information to pass through; when the information is not responsive, the output of GLU is almost zero.

2.4. Mamba block

In the lithium-ion battery RUL prediction task, battery capacity is influenced by environmental temperature, voltage, current, internal resistance, and other factors, constantly evolving in a complex state. Therefore, the problem can be transformed into a high-dimensional state space, and a State Space Model (SSM) can be utilized for lithium-ion battery RUL prediction. In this approach, we use Mamba (Selective State Space Model) [62] as the backbone of our model, which enhances the model's ability to capture continuous-time states.

The discretized SSM model can be formulated in two modes: CNN mode and RNN mode. The computational formula for the RNN mode is shown as follows:

$$x(k+1) = \overline{A}x(k) + \overline{B}u(k) \tag{8a}$$

$$y(k) = \overline{C}x(k) + \overline{D}u(k) \tag{8b}$$

where $\overline{A}$, $\overline{B}$, $\overline{C}$ and $\overline{D}$ are the discrete-time dynamic matrices obtained by discretizing A, B, C, and D using the time step $\Delta \in \mathbb{R}$, which may have complex-valued components. k represents the discrete time index. $u(k)$ is the input at step k, $y(k)$ is the output step k, and $x(k)$ is the hidden state of the model corresponding to the input $u(k)$ at step k. The calculation formulas using zero-order hold method are shown as follows:

$$\overline{A} = exp(\Delta A) \tag{9}$$

$$\overline{B} = (\Delta A)^{-1}(exp(\Delta A) - I)\cdot \Delta B \tag{10}$$

$$\overline{C} = C \text{ and } \overline{D} = D \tag{11}$$

Although learning the dynamics in Eqs. (8a) and (8b) is a major focus of the Mamba block, these dynamics are not simply implemented in isolation. In fact, preprocessing of the input $u(k)$ and post-processing of

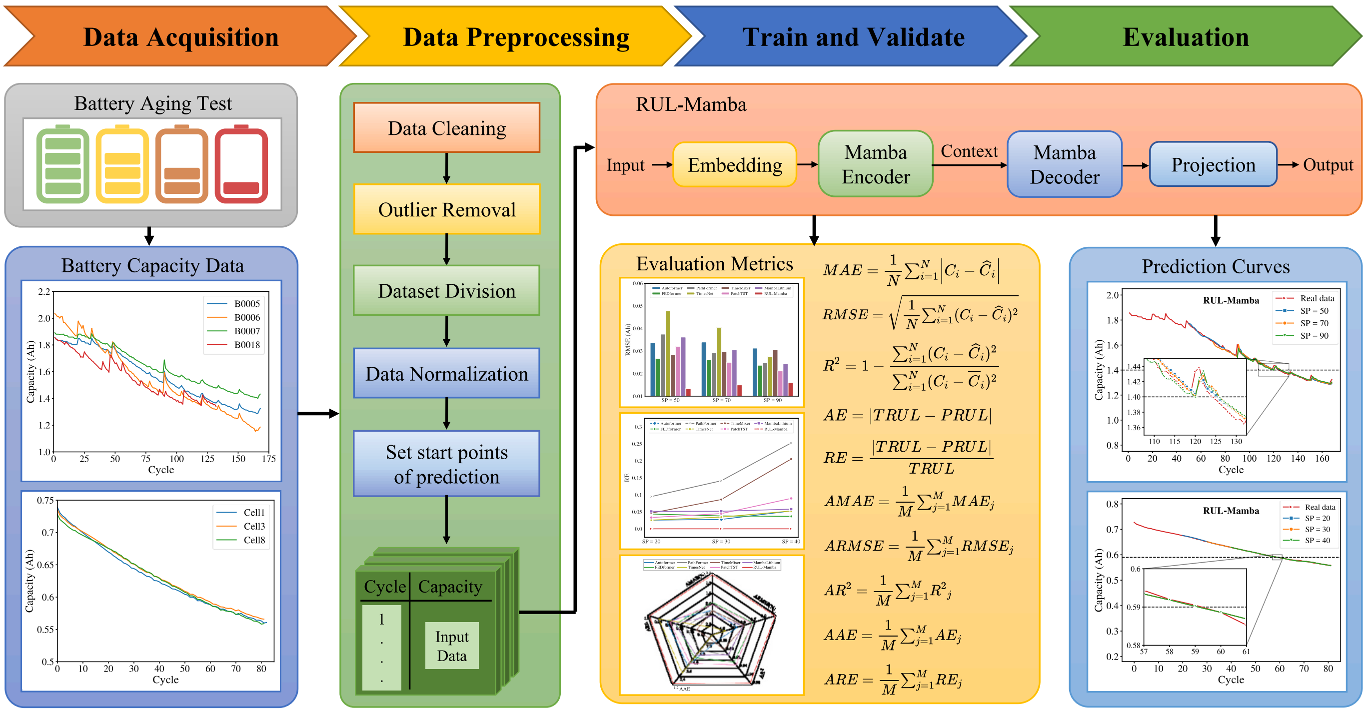


Fig. 2. The flowchart of RUL prediction procedure.

the output $y(k)$ are necessary for ensuring good performance. As shown in Fig. 1(d), it illustrates the specific implementation of the Mamba block in our proposed model. Before feeding the input $\tilde{u}(k)$ to the SSM, perform linear mapping, convolution, and SiLU activation [71] on it to obtain $u(k)$. Once the SSM calculates the output $y(k)$, a multiplication gating operation is performed to control the flow of information from the input $\tilde{u}(k)$ to the output $\tilde{y}(k)$. Intuitively, the gate $g(\tilde{y}, \tilde{u})$ controls which outputs $\tilde{y}$ are set to zero based on the input $\tilde{u}$.

2.5. Loss function

In this study, the training process of the RUL-Mamba model is meticulously orchestrated by minimizing the validation loss, with the loss function being the Symmetric Mean Absolute Percentage Error (SMAPE). The SMAPE function, which serves as a crucial metric for quantifying the disparity between the predicted and actual values, is mathematically formulated as follows:

$$SMAPE = \frac{100\%}{N}\sum_{i=1}^{N}\frac{|y_i - \hat{y}_i|}{(|y_i| + |\hat{y}_i|)/2} \tag{12}$$

where y_i represents the actual observed value, $\hat{y}_i$ denotes the corresponding predicted value generated by the model, and N stands for the total number of samples encompassed within the validation set. The SMAPE is bounded within the range of 0 to 2 and a lower SMAPE value within this range serves as a clear indicator of enhanced prediction accuracy.

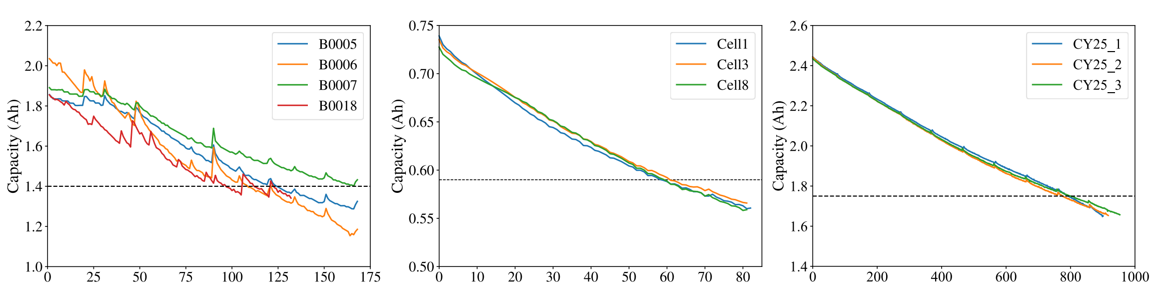
2.6. RUL prediction procedure

Fig. 2 presents the detailed flowchart of our proposed method, which encompasses data acquisition, data preprocessing, model training and validation, as well as model evaluation. The steps involved in the RUL prediction process are summarized as follows:

Step 1: Collect battery capacity degradation data under discharge conditions to construct the dataset.

Step 2: Replace missing values by applying linear interpolation, remove outliers based on the 2-σ principle, divide the cleaned dataset into training dataset and testing dataset, reduce the differences between capacity values through Min-Max normalization, and then set appropriate prediction starting points.

Step 3: Train the RUL-Mamba model by minimizing the validation loss. One battery is used as the testing dataset, and the remaining batteries are used as the training dataset. The first 80 % of the training dataset are used for training, and the remaining 20 % are used for validation.



(a) NASA (b) Oxford (c) TJU

Fig. 3. Capacity degradation curves for lithium-ion batteries in three datasets: (a) NASA dataset, (b) Oxford dataset, (c) TJU dataset.

Table 2
Batteries information of NASA, Oxford and TJU datasets.

Source	Battery ID	Charge/discharge cut-off voltage	Charge/discharge constant current	Temperature	Rated capacity	End-of-life criteria
NASA	B0005	4.2 V/2.7 V	1.5 A/2 A	24 °C	2 Ah	1.40 Ah
	B0006	4.2 V/2.5 V	1.5 A/2 A	24 °C	2 Ah	1.40 Ah
	B0007	4.2 V/2.2 V	1.5 A/2 A	24 °C	2 Ah	1.40 Ah
	B0018	4.2 V/2.5 V	1.5 A/2 A	24 °C	2 Ah	1.40 Ah
Oxford	Cell1	4.2 V/2.7 V	0.74 A/0.74 A	40 °C	0.74 Ah	0.59 Ah
	Cell3	4.2 V/2.7 V	0.74 A/0.74 A	40 °C	0.74 Ah	0.59 Ah
	Cell8	4.2 V/2.7 V	0.74 A/0.74 A	40 °C	0.74 Ah	0.59 Ah
TJU	CY25_1	4.2 V/2.5 V	1.25 A/2.5 A	25 °C	2.5 Ah	1.75 Ah
	CY25_2	4.2 V/2.5 V	1.25 A/2.5 A	25 °C	2.5 Ah	1.75 Ah
	CY25_3	4.2 V/2.5 V	1.25 A/2.5 A	25 °C	2.5 Ah	1.75 Ah

Step 4: Send the testing dataset to the trained model for RUL prediction after the training process. Analyze and visualize the prediction results. Utilize evaluation metrics (MAE, RMSE, R^2, AE and RE) and calculate average metrics (AMAE, ARMSE, AR^2, AAE and ARE) to assess the model's performance on three public datasets.

3. Datasets and experimental settings

3.1. Dataset description

To evaluate the performance of the RUL-Mamba model, three public battery degradation datasets are applied. The capacity degradation curves for lithium-ion batteries in three datasets are illustrated in Fig. 3 and detailed information about these batteries is provided in Table 2.

The first public battery degradation dataset [72] (NASA dataset) is the Randomized Battery Usage Data collected by NASA Ames Prognostics Center of Excellence (PCoE). The battery cells of this dataset are 18,650 Li-cobalt cells and have a rated capacity of 2.0 Ah. In our study, we select four batteries (B0005, B0006, B0007, and B0018) that underwent repeated charging and discharging processes. Each battery was charged at a constant current of 1.5 A until the voltage reached 4.2 V, followed by a constant voltage charge at 4.2 V until the charging current fell below 20 mA. Batteries B5, B6, and B7 were discharged at a constant current of 2 A until their voltages dropped to 2.7 V, 2.5 V, and 2.2 V, respectively, while battery B18 was discharged to 2.5 V. The entire process was conducted at room temperature. Discharge curves for all batteries were recorded every 50 cycles, capturing the discharge capacity, voltage, current, and cell temperature at each step. It is generally accepted that a battery is considered to have reached its end of life (EOL) when its actual usable capacity decreases to 70 %–80 % of its rated capacity [20]. The NASA dataset defines 70 % of its rated capacity as EOL, which is taken as 1.40 Ah in this paper.

The second Oxford dataset [73] is the Oxford Battery Degradation dataset collected by Oxford University. This dataset includes degradation data from eight lithium-ion pouch cells, each with a nominal capacity of 0.74 Ah and a nominal voltage of 4.2 V. In this study, we select three batteries numbered Cell1, Cell3 and Cell8. All batteries were tested under the same drive cycle at 40 °C, continuously charged and discharged until they reached 80 % of their nominal capacity. Characterization measurements were taken every 100 cycles, recording the voltage, current, discharge capacity, and cell temperature at each step. Generally, the Oxford dataset defines 80 % of its rated capacity as EOL, which is considered to be 0.59 Ah.

The third TJU dataset [74] is the commercial lithium-ion battery cycling dataset collected by Tongji University, which consists of batteries with 42 (3) wt% Li(NiCoMn)O2 blended with 58 (3) wt% Li(NiCoAl) O2 positive electrode, known as NCM + NCA batteries, which have a rated capacity of 2.5 Ah. In this study, we select a group of batteries designated as CY25_1, CY25_2, and CY25_3, which were charged and discharged at current rates of 0.5C/1C. The voltage and current data encompass the charging, discharging, and relaxation processes. The battery cycling involved constant current charging at a rate of 0.5C (1.25 A) to 4.2 V, followed by a constant voltage charging step at 4.2 V until the current dropped to 0.05C (125 mA). Subsequently, the NCM + NCA batteries were discharged to 2.5 V at a current rate of 1C (2.5 A). The relaxation time between the constant voltage charging and the constant current discharging for the NCM + NCA battery is set at 60 min, with a sampling interval of 30 s. The TJU dataset defines the EOL of the batteries as 70 % of their rated capacity, which corresponds to 1.75 Ah.

Table 3
Dataset division of NASA, Oxford and TJU datasets.

Dataset	Training dataset	Test dataset
NASA	B0006 B0007 B0018	B0005
Oxford	Cell1 Cell3	Cell8
TJU	CY25_2 CY25_3	CY25_1

3.2. Dataset division

In this paper, the dataset is divided into a training dataset and a testing dataset. For the NASA dataset, batteries B0006, B0007, and B0018 are used for training, while the data from battery B0005 are used for testing. For the Oxford dataset, batteries Cell1 and Cell3 are utilized for training, and battery Cell8 is utilized for testing. For the TJU dataset, batteries CY25_2 and CY25_3 are used for training, and battery CY25_1 is used for testing. The specific dataset division is shown in Table 3. The first 80 % of the training dataset is used for training, and the last 20 % is used for validation. For the testing dataset, the model is evaluated from the respective prediction starting points.

3.3. Hyperparameter settings

Hyperparameter optimization stands as a pivotal determinant that exerts a profound impact on the performance of lithium-ion battery RUL prediction models. In the complex landscape of model tuning, adaptive optimization algorithms emerge as powerful tools, empowered with the ability to unearth the most optimal configuration of network hyperparameters. By doing so, they pave the way for a more rational and highly efficient network architecture, circumventing the common pitfall of models getting ensnared in local optima – an issue that plagues manual parameter tuning efforts. In this study, we employ the Tree-structured Parzen Estimator (TPE) [75] for hyperparameter optimization on the NASA and Oxford datasets, respectively, which is a probabilistic model-based approach provided by the Optuna library. Table 4 lists the key hyperparameter search range and optimal hyperparameter settings for the RUL-Mamba model, encompassing learning rate, dropout rate, and the number of neurons in hidden layers. For performance comparisons, all baseline models are configured with 2 encoder layers and 1 decoder layer (if a decoder is present), and 16 neurons in hidden layers. This is a fair comparison relative to RUL-Mamba's hyperparameters.

To ensure robustness and statistical significance, the number of trials is fixed at 200, with the Adam optimizer being harnessed to drive the

Table 4
Hyperparameter search range and optimal hyperparameter settings of the RUL-Mamba.

	Hyperparameters	Search range	NASA dataset	Oxford dataset	TJU dataset
RUL-Mamba	Number of units in hidden layers	[8, 128]	48	16	16
	Number of decoder layers	[1, 3]	1	2	2
	Learning rate	[1e−4, 1e−2]	0.0022	0.001	0.001
	Dropout rate	[0.01, 0.2]	0.0615	0.1	0.1

training process forward. For every individual model, both the training and testing regimens on each dataset is replicated 10 times, and the ensuing test results is averaged across these ten experimental runs, thereby enhancing the reliability and generalizability of the findings. Besides, to accelerate the training process and reduce computational overhead, we introduce an early stopping mechanism. All experiments are carried out on a workstation armed with dual Intel (R) Xeon (R) E5-2643 v4 CPUs @ 3.4 GHz (12 cores) and 8 NVIDIA TESLA V100 GPUs.

3.4. Evaluation metrics

In this study, we use five evaluation metrics to assess the model's prediction performance for each starting point (SP): Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), R-squared (R^2), Absolute Error (AE), and Relative Error (RE). The calculation formulas for these metrics are as follows:

$$MAE = \frac{1}{N}\sum_{i=1}^{N}\left|C_i - \widehat{C}_i\right| \tag{13}$$

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(C_i - \widehat{C}_i\right)^2} \tag{14}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N}\left(C_i - \widehat{C}_i\right)^2}{\sum_{i=1}^{N}\left(C_i - \overline{C}_i\right)^2} \tag{15}$$

$$AE = |TRUL - PRUL| \tag{16}$$

$$RE = \frac{|TRUL - PRUL|}{TRUL} \tag{17}$$

where C_i symbolizes the actual capacity value, $\overline{C}_i$ represents the mean of the actual battery capacity values, $\widehat{C}_i$ denotes the predicted capacity value, *PRUL* stands for the predicted value of RUL, and *TRUL* indicates the true value of RUL. For MAE, RMSE, AE, and RE, closer values to zero indicate higher RUL prediction accuracy. Conversely, a higher R^2 value signifies more accurate RUL prediction results.

To distill a more holistic assessment, we also average the aforementioned metrics across three different prediction SPs to derive five additional evaluation metrics: Average Mean Absolute Error (AMAE), Average Root Mean Squared Error (ARMSE), Average R-squared (AR^2), Average Absolute Error (AAE), and Average Relative Error (ARE). This multi-faceted evaluation framework not only facilitates a granular understanding of the model's performance at individual prediction starting points but also enables a macroscopic view through the aggregated metrics, endowing us with a more comprehensive and nuanced insight into the overall predictive capabilities of the RUL-Mamba model.

4. Experimental results and discussions

4.1. Comparison with SOTA TSF methods

To fully verify the performance of the method proposed in this paper, the proposed method is compared with several representative TSF methods [61,76–81] on the NASA, Oxford and TJU datasets. Autoformer [76] is a Transformer-based long-term forecasting model that distinguishes long-term trend information and captures subsequence similarities through a decomposition architecture featuring an autocorrelation mechanism. FEDformer [77] introduces Fourier-enhanced blocks and Wavelet-enhanced blocks into the Transformer architecture to capture key structures in time series data through frequency domain mapping. PathFormer [78] implements adaptive multi-scale feature extraction of temporal signals based on the Pathways architecture, building upon the Transformer framework. TimesNet [79] extends one-dimensional time series into a two-dimensional space for analysis and captures changes across cycles and within cycles using a parameter-efficient Inception block [82] in this two-dimensional space. TimeMixer [80], based on a fully multilayer perceptron (MLP) architecture, effectively decouples and integrates seasonal and trend features of complex time series at different scales. PatchTST [81] proposes a channel-independent patch time series Transformer. MambaLithium [61] is a simplified implementation of Mamba [62], representing an innovative linear time series modeling approach that merges the step-wise processing capabilities of recurrent neural networks (RNNs) with the global information processing capabilities of convolutional neural networks (CNNs) within a SSM framework. Extensive comparisons with these aforementioned methods are conducted in this paper to evaluate the effectiveness of the proposed method.

4.1.1. RUL prediction on NASA dataset

In order to conduct a comprehensive assessment of the RUL prediction performance exhibited by diverse methods on the NASA dataset, we carry out RUL predictions with starting points set at 50, 70, and 90 (SP = 50, 70, 90), respectively. These carefully chosen SPs allow us to scrutinize the models' performance across different phases of the battery's lifecycle. As shown in Fig. 4(a)–(c), under the identical SP condition, the capacity prediction curve generated by RUL-Mamba demonstrates a remarkable proximity to the actual capacity degradation curve when compared to those of the other seven models in contention. This closer alignment is a testament to the superior predictive capabilities of RUL-Mamba, which is adept at tracing the fluctuations and trends of battery capacity. Moreover, upon examining the local magnification of the capacity degradation curve at the EOL stage, it becomes strikingly evident that RUL-Mamba outperforms its counterparts in a rather nuanced aspect. Not only does it achieve the optimal prediction performance throughout the normal aging region of the battery, but it also showcases an impressive ability to effectively capture and precisely fit the local capacity regeneration phenomena.

A thorough analysis of the various evaluation metrics presented in Fig. 5(a)–(f) leads to the following conclusions: a) Across a spectrum of SP conditions, RUL-Mamba unfailingly exhibits remarkable superiority. It manages to secure the lowest values for MAE, RMSE, AE, and RE, which are crucial indicators of prediction accuracy. Simultaneously, it attains the highest R^2 value, a metric that powerfully reflects the degree of fit between the predicted and actual values. b) It is challenging to determine the performance ranking of other baseline models on the NASA dataset under the same SP conditions. Some models perform well in predicting the actual capacity degradation curve (MAE and RMSE) but may perform poorly in predicting true RUL values (AE and RE). For example, under the identical SP conditions, the MAE and RMSE metrics of FEDformer are better than those of TimesNet, while its AE and RE metrics are worse. In contrast, RUL-Mamba outperforms all other baseline models across all metrics. c) The performance metrics of different models are subject to considerable fluctuations as the SP conditions vary, illustrating that the selection of prediction SP can influence

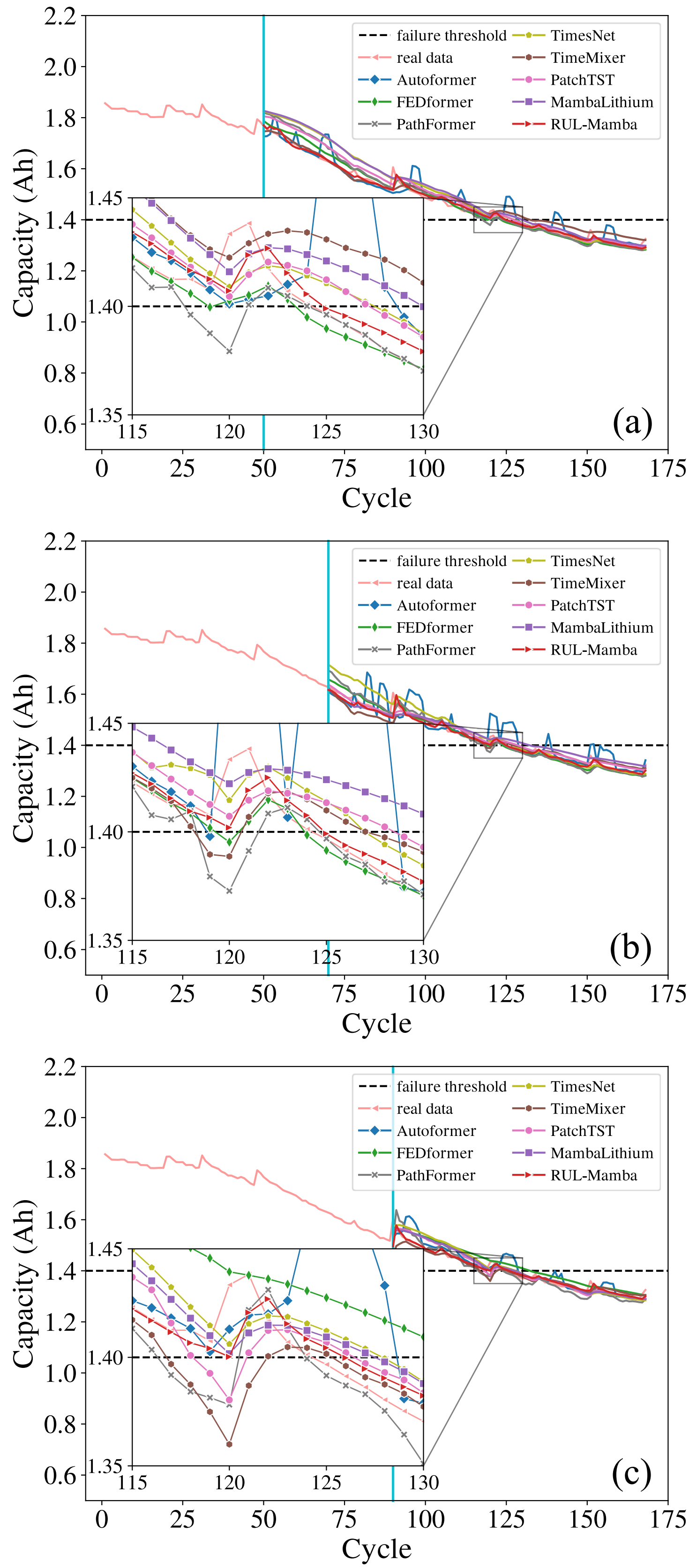


Fig. 4. Comparison of RUL prediction curves from different methods for NASA dataset. (a) prediction curves for SP = 50; (b) prediction curves for SP = 70; (c) prediction curves for SP = 90.

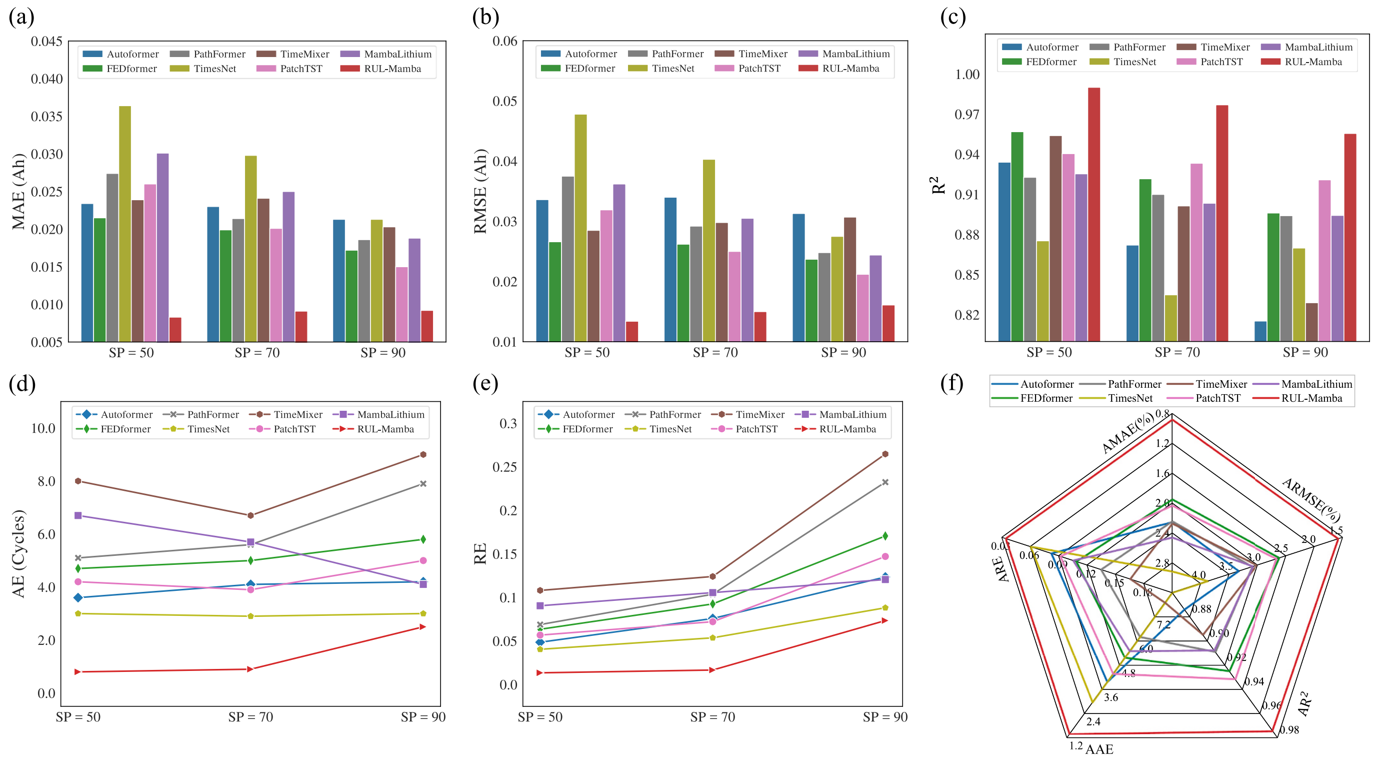


Fig. 5. Evaluation metrics of different methods for NASA dataset. (a) MAE. (b) RMSE. (c) R^2. (d) AE. (e) RE. (f) AMAE, ARMSE, AR^2, AAE, ARE.

Table 5
RUL prediction results of different models on NASA dataset. SP: starting point of prediction; TRUL: average value of true RUL; PRUL: average value of predicted RUL; AE: average absolute error of 10 experiments.

Dataset	Method	SP	TRUL	PRUL	MAE	RMSE	R^2	AE	RE	AMAE	ARMSE	AR^2	AAE	ARE
NASA	Autoformer	50	75	74.8	0.0234	0.0336	0.9341	3.6	0.0486	0.0226	0.0330	0.8738	4.0	0.0827
		70	55	52.3	0.0230	0.0340	0.8721	4.1	0.0759					
		90	35	32.8	0.0213	0.0313	0.8153	4.2	0.1235					
	FEDformer	50	75	74.9	0.0215	0.0266	0.9569	4.7	0.0635	0.0195	0.0255	0.9249	5.2	0.1089
		70	55	53.6	0.0199	0.0262	0.9217	5.0	0.0926					
		90	35	32.6	0.0172	0.0237	0.8961	5.8	0.1706					
	PathFormer	50	75	70.3	0.0274	0.0375	0.9228	5.1	0.0689	0.0225	0.0305	0.9090	6.2	0.1350
		70	55	50.0	0.0214	0.0292	0.9100	5.6	0.1037					
		90	35	27.1	0.0186	0.0248	0.8941	7.9	0.2324					
	TimesNet	50	75	78.0	0.0364	0.0478	0.8753	3.0	0.0405	0.0292	0.0385	0.8600	3.0	0.0608
		70	55	57.9	0.0298	0.0403	0.8349	2.9	0.0537					
		90	35	38.0	0.0213	0.0275	0.8699	3.0	0.0882					
	TimeMixer	50	75	79.8	0.0239	0.0285	0.9540	8.0	0.1081	0.0228	0.0297	0.8948	7.9	0.1656
		70	55	52.7	0.0241	0.0298	0.9014	6.7	0.1241					
		90	35	26.0	0.0203	0.0307	0.8290	9.0	0.2647					
	PatchTST	50	75	79.2	0.0260	0.0319	0.9405	4.2	0.0568	0.0204	0.0260	0.9316	4.4	0.0920
		70	55	56.7	0.0201	0.0250	0.9333	3.9	0.0722					
		90	35	30.4	0.0150	0.0212	0.9209	5.0	0.1471					
	MambaLithium	50	75	81.7	0.0301	0.0362	0.9254	6.7	0.0905	0.0246	0.0304	0.9077	5.5	0.1056
		70	55	60.7	0.0250	0.0305	0.9034	5.7	0.1056					
		90	35	32.9	0.0188	0.0244	0.8943	4.1	0.1206					
	RUL-Mamba	50	75	75.8	**0.0083**	**0.0134**	**0.9901**	**0.8**	**0.0135**	**0.0089**	**0.0148**	**0.9742**	**1.4**	**0.0346**
		70	55	55.9	**0.0091**	**0.0150**	**0.9770**	**0.9**	**0.0167**					
		90	35	34.5	**0.0092**	**0.0161**	**0.9556**	**2.5**	**0.0735**					

The best results are highlighted in bold.

RUL prediction performance. In comparison, RUL-Mamba's RUL prediction results are less sensitive to changes in the SP, achieving excellent prediction performance across all conditions. d) When considering all five statistical average metrics, RUL-Mamba reigns supreme, occupying the top position without equivocation.

Table 5 displays the evaluation metrics of eight methods on the NASA dataset under different prediction SP conditions. The results show that the evaluation metrics of the proposed method, namely MAE, RMSE, AE, RE and R^2, remain the best under different SP conditions all the time, and its five statistical average metrics are also the optimal. This further proves that the proposed method can effectively improve the accuracy of RUL prediction.

4.1.2. RUL prediction on Oxford dataset

Compared to the NASA dataset, the Oxford dataset has a shorter capacity degradation sequence and does not exhibit local capacity

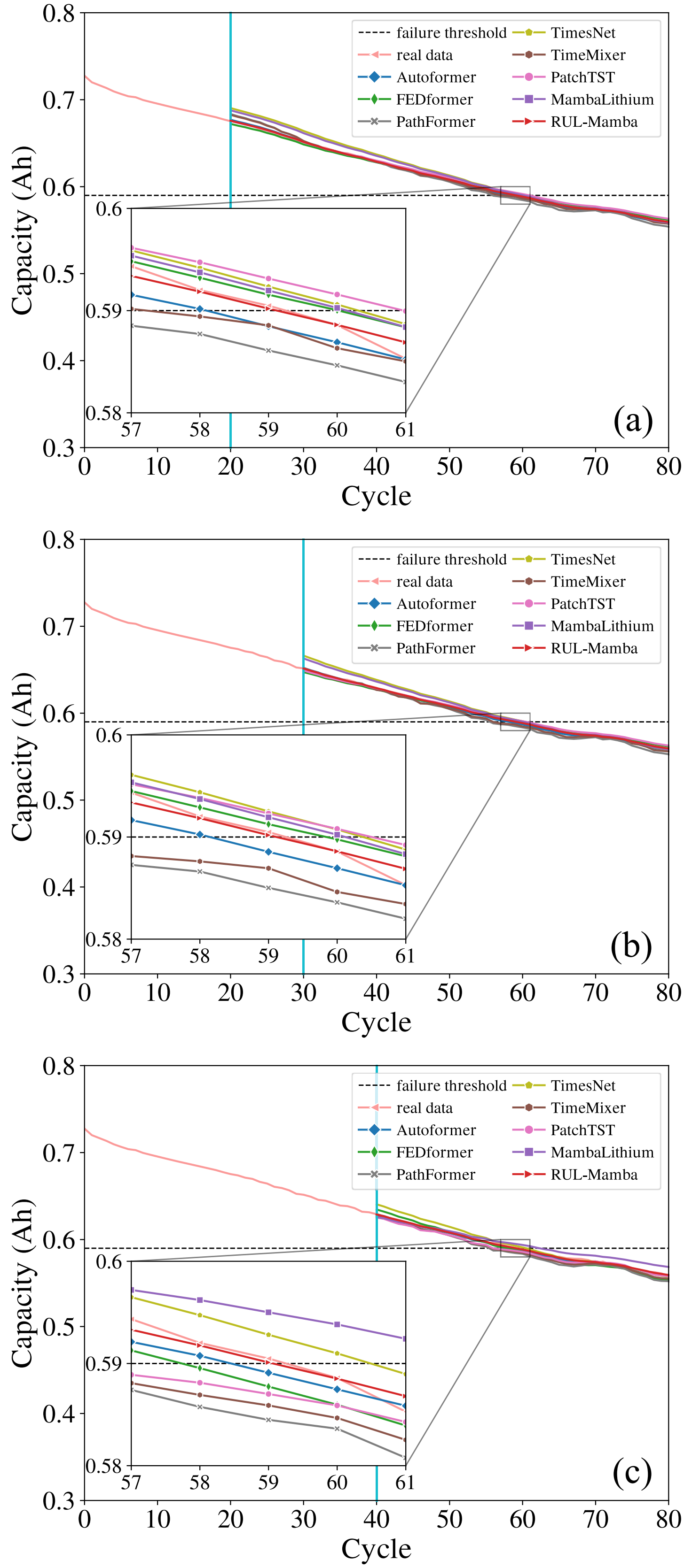


Fig. 6. Comparison of RUL prediction curves from different methods for Oxford dataset. (a) prediction curves for SP = 20; (b) prediction curves for SP = 30; (c) prediction curves for SP = 40.

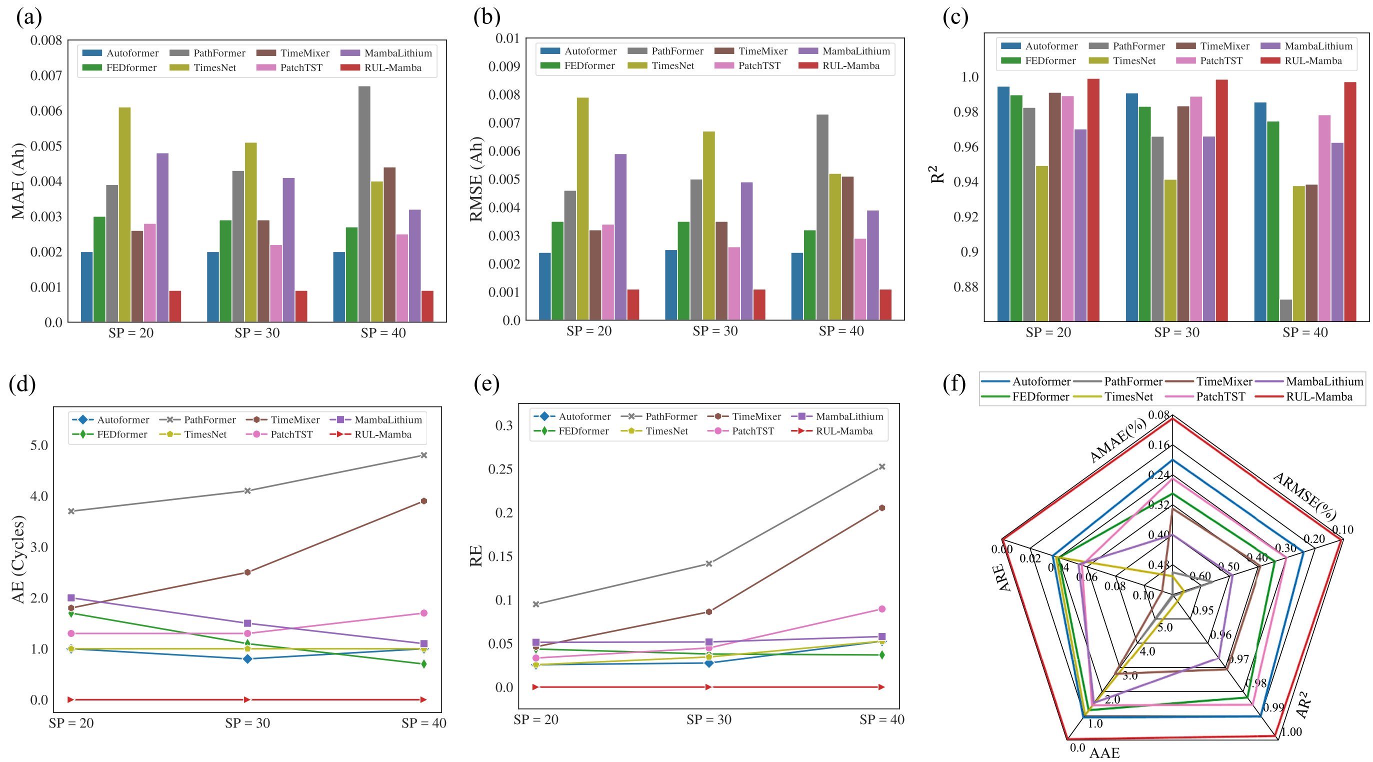


Fig. 7. Evaluation metrics of different methods for Oxford dataset. (a) MAE. (b) RMSE. (c) R^2. (d) AE. (e) RE. (f) AMAE, ARMSE, AR^2, AAE, ARE.

Table 6
RUL prediction results of different models on Oxford dataset. SP: starting point of prediction; TRUL: average value of true RUL; PRUL: average value of predicted RUL; AE: average absolute error of 10 experiments.

Dataset	Method	SP	TRUL	PRUL	MAE	RMSE	R^2	AE	RE	AMAE	ARMSE	AR^2	AAE	ARE
Oxford	Autoformer	20	40	39.2	0.0020	0.0024	0.9945	1.0	0.0256	0.0020	0.0024	0.9902	0.9	0.0353
		30	30	29.2	0.0020	0.0025	0.9907	0.8	0.0276					
		40	20	19.0	0.0020	0.0024	0.9855	1.0	0.0526					
	FEDformer	20	40	39.5	0.0030	0.0035	0.9896	1.7	0.0436	0.0029	0.0034	0.9824	1.2	0.0394
		30	30	29.5	0.0029	0.0035	0.9830	1.1	0.0379					
		40	20	19.3	0.0027	0.0032	0.9746	0.7	0.0368					
	PathFormer	20	40	36.3	0.0039	0.0046	0.9824	3.7	0.0949	0.0050	0.0056	0.9404	4.2	0.1630
		30	30	25.9	0.0043	0.0050	0.9659	4.1	0.1414					
		40	20	15.2	0.0067	0.0073	0.8728	4.8	0.2526					
	TimesNet	20	40	41.0	0.0061	0.0079	0.9492	1.0	0.0256	0.0051	0.0066	0.9427	1.0	0.0376
		30	30	31.0	0.0051	0.0067	0.9413	1.0	0.0345					
		40	20	21.0	0.0040	0.0052	0.9377	1.0	0.0526					
	TimeMixer	20	40	39.0	0.0026	0.0032	0.9910	1.8	0.0462	0.0033	0.0039	0.9709	2.7	0.1126
		30	30	27.9	0.0029	0.0035	0.9833	2.5	0.0862					
		40	20	16.1	0.0044	0.0051	0.9385	3.9	0.2053					
	PatchTST	20	40	41.1	0.0028	0.0034	0.9891	1.3	0.0333	0.0025	0.0030	0.9854	1.4	0.0559
		30	30	30.3	0.0022	0.0026	0.9888	1.3	0.0448					
		40	20	18.9	0.0025	0.0029	0.9782	1.7	0.0895					
	MambaLithium	20	40	42.0	0.0048	0.0059	0.9701	2.0	0.0513	0.0040	0.0049	0.9662	1.5	0.0536
		30	30	31.5	0.0041	0.0049	0.9660	1.5	0.0517					
		40	20	21.1	0.0032	0.0039	0.9624	1.1	0.0579					
	RUL-Mamba	20	40	40.0	**0.0009**	**0.0011**	**0.9990**	**0.0**	**0.0000**	**0.0009**	**0.0011**	**0.9982**	**0.0**	**0.0000**
		30	30	30.0	**0.0009**	**0.0011**	**0.9985**	**0.0**	**0.0000**					
		40	20	20.0	**0.0009**	**0.0011**	**0.9971**	**0.0**	**0.0000**					

The best results are highlighted in bold.

regeneration phenomena. Therefore, the RUL prediction SPs are set to 20, 30, and 40. As shown in Fig. 6(a)–(c), under the identical SP condition, the capacity prediction curve generated by RUL-Mamba exhibits a significantly closer alignment with the actual capacity degradation curve in comparison to those yielded by the other seven models. Additionally, the local magnification of the capacity degradation curve indicates that RUL-Mamba effectively predicts the capacity degradation trend of normally aging batteries and accurately forecasts the cycles at which battery life terminates, thereby leading to precise RUL prediction.

A detailed analysis of the various evaluation metrics presented in Fig. 7(a)–(f) leads to the following conclusions: a) RUL-Mamba consistently maintains the best RUL prediction performance across all SP conditions, achieving the lowest metrics for MAE, RMSE, AE, and RE while consistently attaining the highest R^2 value. Especially, RUL-Mamba shows a prediction error (AE) of zero cycles for TRUL. b) The performance metrics of different models fluctuate with changes in SP,

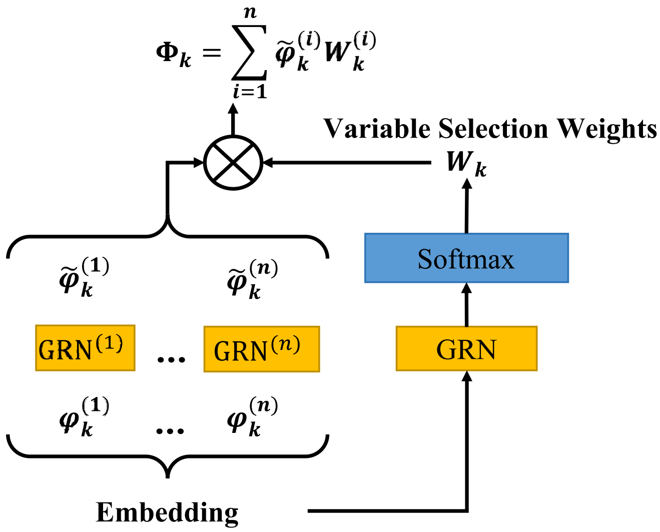


Fig. 8. The architecture of Variable Attention Network (VAN).

indicating that the choice of prediction SP can impact the models' predictive performance. In contrast, RUL-Mamba's RUL prediction results are least affected by variations in the SP. c) As SP increases, the R^2 metrics of all models decrease, while the trends of other metrics vary depending on the model. d) RUL-Mamba ranks highest across all five statistical average metrics.

Table 6 shows the evaluation metrics of eight methods on the Oxford dataset under different SP conditions. The results indicate that the evaluation metrics MAE, RMSE, AE, RE and R^2 of the proposed method are superior to those of the other seven methods under different SP conditions, and its five statistical average metrics are also the best. This further proves that the proposed method can significantly improve the accuracy of RUL prediction.

4.1.3. *Supplementary experiments on TJU dataset*

In our study on the NASA and Oxford datasets, we investigate RUL prediction using battery capacity as a single-variable input. The analysis of the results indicate that the RUL-Mamba model demonstrates optimal performance for RUL prediction based on a single variable. However, utilizing multiple battery health indicators (HIs) for multivariate time series forecasting is also a common technical approach for RUL prediction. This section presents supplementary experiments conducted on the TJU dataset to evaluate the effectiveness of the RUL-Mamba model in RUL prediction with multivariate inputs.

Following the feature extraction method proposed by Wang et al. [83], we select various health indicators, including the mean, standard deviation, kurtosis, skewness, charging time, accumulated charge, curve slope, and curve entropy derived from the selected current and voltage curves, resulting in a total of 16 HIs. The input variable dimension for the model is set to 17 (including capacity), meaning we predict future capacity using historical capacity along with the 16 HIs, thereby facilitating RUL prediction. Since the RUL-Mamba model architecture is originally designed for single-variable time series forecasting, we introduce a Variable Attention Network (VAN) [31] (as shown in Fig. 8) before the FAN in its encoder to effectively model the dependencies among multiple variables, thus representing the new model as RUL-Mamba*.

Compared to the NASA and Oxford datasets, the capacity degradation data collected in the TJU dataset features smoother actual capacity degradation curves, with a lower frequency and smaller magnitude of capacity recovery phenomena. This makes it easier for the model to predict the actual capacity degradation curve, resulting in lower MAE and RMSE values, as well as a higher R^2. As shown in Table 7 and Fig. 10, RUL-Mamba* model outperforms the TimesNet model across most evaluation metrics, only slightly lagging behind in the AE and RE metrics. However, TimesNet ranks lowest among the eight models in other metrics, demonstrating overall inferior performance compared to RUL-Mamba*. Besides, the prediction performance of RUL-Mamba* is less affected by prediction SP. Fig. 9 illustrates that under different SP conditions, the RUL-Mamba* model provided the best fit for the actual capacity degradation curve. The experimental results on the TJU dataset confirm the effectiveness of RUL-Mamba* in RUL prediction with multivariate inputs, further validating the reliability and robustness of the RUL-Mamba framework, and providing strong support for its broad application in real-world battery systems.

Table 7
RUL prediction results of different models with multivariate inputs on TJU dataset. SP: starting point of prediction; TRUL: average value of true RUL; PRUL: average value of predicted RUL; AE: average absolute error of 10 experiments.

Dataset	Method	SP	TRUL	PRUL	MAE	RMSE	R^2	AE	RE	AMAE	ARMSE	AR^2	AAE	ARE
TJU	Autoformer	200	579	580.4	0.0020	0.0028	0.9997	1.6	0.0028	0.0030	0.0040	0.9987	3.1	0.0070
		300	479	478.6	0.0031	0.0042	0.9989	3.2	0.0067					
		400	379	376.6	0.0038	0.0051	0.9975	4.4	0.0116					
	FEDformer	200	579	581.2	0.0020	0.0028	0.9997	2.2	0.0038	0.0028	0.0037	0.9989	2.6	0.0058
		300	479	481.5	0.0030	0.0039	0.9990	2.5	0.0052					
		400	379	380.2	0.0034	0.0045	0.9980	3.2	0.0085					
	PathFormer	200	579	569.2	0.0093	0.0123	0.9938	9.8	0.0170	0.0114	0.0157	0.9816	12.0	0.0266
		300	479	467.4	0.0105	0.0148	0.9868	11.6	0.0243					
		400	379	364.4	0.0144	0.0199	0.9641	14.6	0.0386					
	TimesNet	200	579	580.0	0.0161	0.0202	0.9832	**1.0**	**0.0017**	0.0145	0.0178	0.9810	**0.6**	**0.0013**
		300	479	479.1	0.0145	0.0177	0.9812	**0.1**	**0.0002**					
		400	379	378.3	0.0129	0.0154	0.9787	**0.7**	**0.0019**					
	TimeMixer	200	579	582.2	0.0120	0.0147	0.9900	12.0	0.0208	0.0195	0.0230	0.9507	32.8	0.0767
		300	479	456.3	0.0195	0.0236	0.9630	35.1	0.0734					
		400	379	334.4	0.0271	0.0308	0.8992	51.4	0.1360					
	PatchTST	200	579	574.8	0.0087	0.0113	0.9947	8.8	0.0152	0.0206	0.0231	0.9442	25.2	0.0599
		300	479	456.6	0.0206	0.0233	0.9640	22.8	0.0477					
		400	379	334.9	0.0326	0.0347	0.8740	44.1	0.1167					
	MambaLithium	200	579	580.9	0.0064	0.0082	0.9970	4.7	0.0081	0.0110	0.0130	0.9840	12.0	0.0291
		300	479	472.3	0.0088	0.0110	0.9919	6.7	0.0140					
		400	379	354.3	0.0178	0.0197	0.9631	24.7	0.0653					
	RUL-Mamba*	200	579	581.6	**0.0014**	**0.0022**	**0.9998**	2.6	0.0045	**0.0015**	**0.0023**	**0.9997**	2.6	0.0056
		300	479	481.6	**0.0015**	**0.0023**	**0.9997**	2.6	0.0054					
		400	379	381.6	**0.0016**	**0.0024**	**0.9995**	2.6	0.0069					

The best results are highlighted in bold.

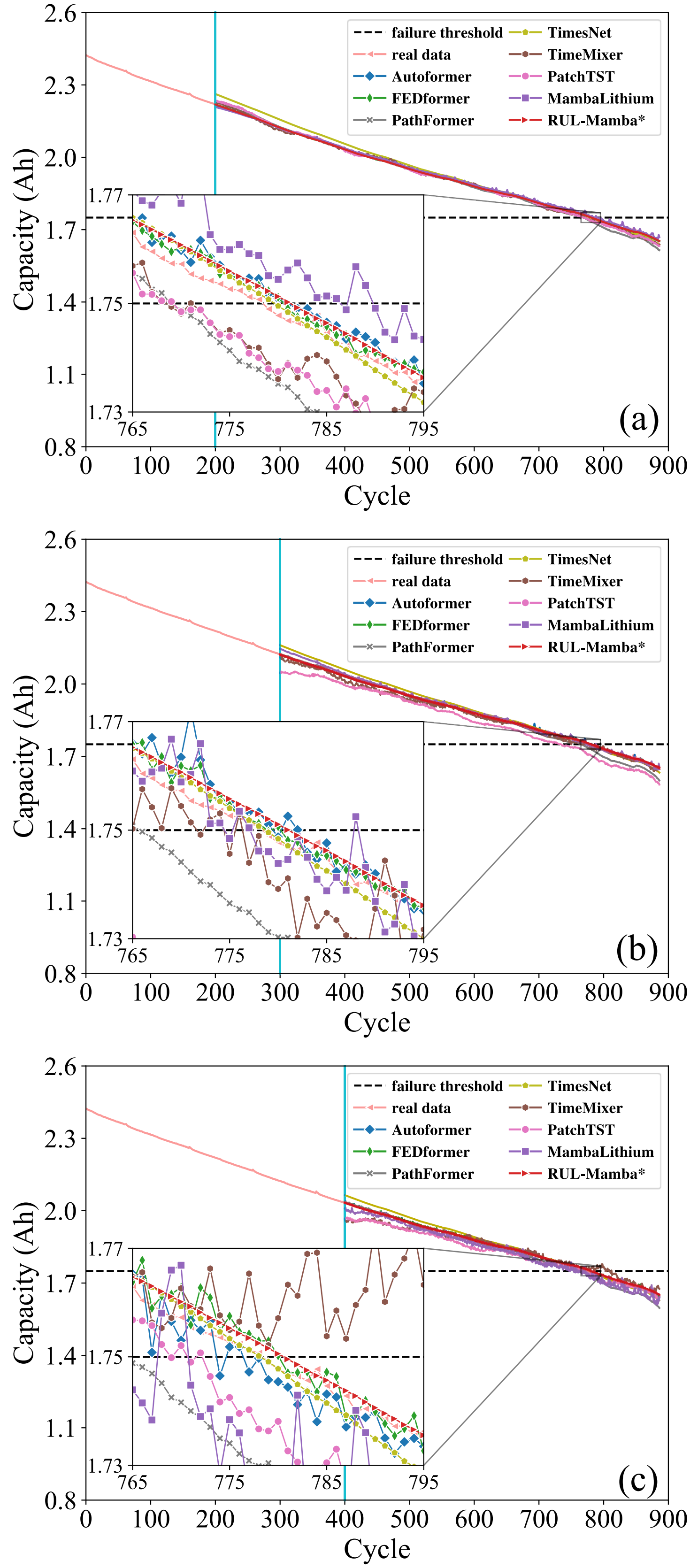


Fig. 9. Comparison of RUL prediction curves from different methods for TJU dataset. (a) prediction curves for SP = 200; (b) prediction curves for SP = 300; (c) prediction curves for SP = 400.

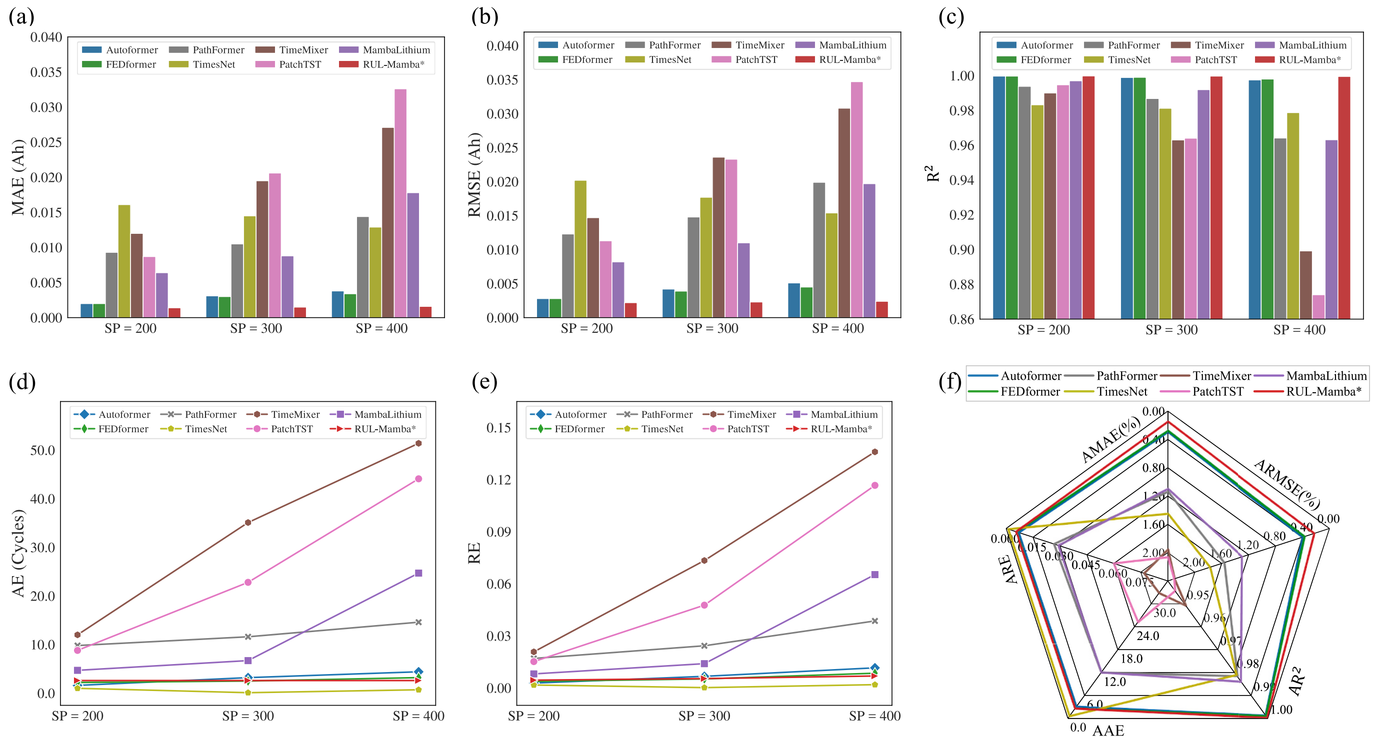


Fig. 10. Evaluation metrics of different methods for TJU dataset. (a) MAE. (b) RMSE. (c) R^2. (d) AE. (e) RE. (f) AMAE, ARMSE, AR^2, AAE, ARE.

Table 8
Computational cost of different methods for TJU dataset.

Method	Training time (s)	Inference time (s)
Autoformer	427.993	0.855
FEDformer	532.561	0.850
PathFormer	111.185	1.375
TimesNet	290.075	0.785
TimeMixer	61.848	0.623
PatchTST	**44.974**	**0.571**
MambaLithium	121.211	0.659
RUL-Mamba*	91.340	0.664

The best results are highlighted in bold.

4.1.4. Comparison of computational cost

Table 8 presents the average time taken for 10 rounds of model training and inference when SP is set at 200, under identical hardware conditions across eight different methods. As shown in Table 8, the average training and inference time of the RUL-Mamba* model is at a lower level. It is worth noting that in the case of similar model complexity, the computational cost of the RUL-Mamba* model is significantly less than that of the Transformer-based models (including Autoformer, FEDformer, and PathFormer). This indicates that the RUL-Mamba* model can achieve efficient RUL prediction for lithium-ion batteries and strongly demonstrates Mamba's capability for high training efficiency and rapid inference speed compared with Transformer.

4.2. Comparison with published RUL prediction methods

The experiments on NASA and Oxford datasets demonstrate that the proposed model can effectively predict battery capacity sequences. We compare our experimental results with other published articles that use similar experimental settings, analyzing the results obtained with almost identical prediction SPs, which further confirms the effectiveness and superiority of our model.

Table 9 summarizes the RUL prediction results for multi-battery experiments on NASA dataset with B0005 as the test battery type under different methods. The results show that, despite some differences in SPs, the proposed model has lower MAE and RMSE values compared to other methods, and it also has lower AE and RE values in most cases.

Table 9
Comparison with published RUL prediction methods on NASA batteries testing dataset.

Battery	Method	Published year	SP	MAE	RMSE	AE	RE
B0005	CNN-ASTLSTM [56]	2022	50	–	0.0183	5	0.0667
			70	–	0.0172	4	0.0727
			90	–	0.0176	**0**	**0.0000**
	PSO multi-kernel RVM [84]	2022	40	0.0258	0.0351	2	0.0290
			60	0.0234	0.0309	1	0.0210
			80	0.0266	0.0348	6	0.2140
	TCN-LSTM [58]	2023	50	0.0152	0.0203	1	0.0133
			70	0.0239	0.0261	8	0.1455
			90	0.0130	0.0172	**0**	**0.0000**
	RUL-Mamba (Ours)	–	50	**0.0083**	**0.0134**	**0.8**	**0.0135**
			70	**0.0091**	**0.0150**	**0.9**	**0.0167**
			90	**0.0092**	**0.0161**	2.5	0.0735

The best results are highlighted in bold.

Table 10
Comparison with published RUL prediction methods on Oxford batteries testing dataset.

Battery	Method	Published year	SP	MAE	RMSE	AE	RE
Cell1	Hybrid-kernel RVM [85]	2022	20	0.0056	0.0057	1	0.0294
Cell2			20	0.0062	0.0082	2	0.0769
Cell3			20	0.0057	0.0530	0	0.0000
Cell8	TCN-LSTM [58]	2023	20	0.0011	0.0018	2	0.0588
			30	**0.0008**	0.0015	1	0.0417
			40	**0.0008**	0.0014	0	0.0000
Cell8	RUL-Mamba (Ours)	–	20	**0.0009**	**0.0011**	**0**	**0.0000**
			30	0.0009	**0.0011**	**0**	**0.0000**
			40	0.0009	**0.0011**	**0**	**0.0000**

The best results are highlighted in bold.

Table 11
RMSE improvement for our experimental results.

Battery	SP	Method	RMSE	RMSE improvement
B0005	50	RUL-Mamba (Ours)	0.0134	33.99 %
		TCN-LSTM [58]	0.0203	
	70	RUL-Mamba (Ours)	0.0150	42.53 %
		TCN-LSTM [58]	0.0261	
	90	RUL-Mamba (Ours)	0.0161	6.40 %
		TCN-LSTM [58]	0.0172	
Cell8	20	RUL-Mamba (Ours)	0.0011	38.89 %
		TCN-LSTM [58]	0.0018	
	30	RUL-Mamba (Ours)	0.0011	26.67 %
		TCN-LSTM [58]	0.0015	
	40	RUL-Mamba (Ours)	0.0011	21.43 %
		TCN-LSTM [58]	0.0014	

In addition, our model is insensitive to changes of prediction SPs and has strong generalization ability.

Table 10 also summarizes the RUL prediction results on Oxford dataset under different methods. The comparative models all use a single battery for testing and multiple other batteries for training. Although there are some differences in SPs and the test battery types, our proposed model shows higher effectiveness and reliability compared to other models. It is worth emphasizing that the prediction error for TRUL (AE) of our model on the Oxford dataset is always zero cycle.

Taking the RMSE results as an example, Table 11 shows the improvement in RMSE in our experimental results. We compare our RMSE results with the best results in published articles. Our experimental results reduce the RMSE by 6.40 %–42.53 % in NASA lithium-ion batteries and by 21.43 %–38.89 % in Oxford lithium-ion batteries.

The above experiments confirm that compared to the models in published articles, the RUL-Mamba model we propose can achieve better prediction performance. Therefore, we can conclude that the proposed model can achieve higher lithium-ion battery RUL prediction accuracy and strong effectiveness and reliability.

5. Conclusion

To improve the accuracy of lithium-ion battery RUL prediction, this paper presents a novel hybrid RUL prediction model (RUL-Mamba) based on Mamba-FAN-GRN, which employs an encoder-decoder architecture. In the encoder, the FAN network and the Mamba block are effectively combined. By utilizing the feature attention mechanism of the FAN network, the effective extraction of key features in each time step is achieved, so that the Mamba blocks can effectively filter and retain the information related to capacity regeneration in the input and discard irrelevant information through the time-varying selection mechanism. In the decoder, the Mamba blocks and the GRN network are effectively combined. By taking advantage of the gating mechanism of the GRN network, the degree of nonlinear processing required for the decoded features output by the Mamba blocks is adaptively controlled, so that the projection layer can accurately model the mapping relationship between the decoded feature vectors and the prediction target. In addition, the optimal hyperparameter settings of the model are obtained through the TPE algorithm, further enhancing the model's performance.

To thoroughly evaluate the performance of RUL-Mamba, we compare it with several representative baseline models, specifically including seven TSF models: Autoformer, FEDformer, PathFormer, TimesNet, TimeMixer, PatchTST, and MambaLithium. The performance of RUL-Mamba is validated across multiple publicly available datasets, leading to the following conclusions:

(1) On the NASA and Oxford datasets, RUL-Mamba achieves the minimum values for MAE, RMSE, AE, and RE, and the maximum R^2 compared to existing superior TSF models under various SP conditions. The associated statistical average metrics—AMAE, ARMSE, AR^2, AAE, and ARE—also showcase optimal performance. This comprehensive achievement not only underlines our model's ability in accurately predicting the actual capacity degradation curve but also highlights its proficiency in precisely forecasting the true value of the RUL, denoted as TRUL.

(2) RUL-Mamba not only has the best performance in RUL prediction with univariate input, but also exhibit robustness and effectiveness in RUL prediction with multivariate inputs. On the TJU dataset, RUL-Mamba* (based on Mamba-VAN-FAN-GRN) shows excellent performance in RUL prediction with multivariate inputs. Additionally, RUL-Mamba exhibits efficient training, fast inference and minimal sensitivity to changes in the prediction SP.

(3) Compared with published RUL prediction methods, the proposed RUL-Mamba model reduces the RMSE by 6.40 %–42.53 % in NASA lithium-ion batteries and by 21.43 %–38.89 % in Oxford lithium-ion batteries, further exhibiting the effectiveness and superiority of our model.

Overall, the RUL-Mamba model has demonstrated its effectiveness and superiority by providing efficient and accurate RUL prediction and skillfully capturing the intricate trends of capacity regeneration and degradation within battery capacity sequences. Looking ahead, our future work will focus on augmenting the model's capabilities by exploring ways to incorporate factors like temperature and current to enhance its real-world applicability, as they're crucial for battery performance and lifespan. Also, we're contemplating combining data-driven with physicochemical models of lithium-ion batteries to create a more robust one for handling RUL and capacity prediction challenges.

CRediT authorship contribution statement

Jiahui Huang: Writing – original draft, Visualization, Software, Methodology, Data curation, Conceptualization. **Lei Liu:** Writing – review & editing, Validation, Supervision, Funding acquisition, Conceptualization. **Hongwei Zhao:** Writing – review & editing, Validation, Investigation, Funding acquisition. **Tianqi Li:** Validation, Supervision. **Bin Li:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data is openly accessible and available for utilization.

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